

Jarvis AI Agent

Phase 4 Architecture Upgrade — Monitoring Signals

Don't upgrade on a schedule — upgrade on signals. This reference card defines exactly when to evolve from a monolithic Agent Core to a hierarchical multi-agent architecture. Log the metrics below from Day 1 so you have baseline data when the time comes.

■ HARD SIGNALS — Upgrade Immediately

1. Context Window Thrashing

The agent starts truncating memory or dropping tool results mid-task. Visible symptom: it 'forgets' something told earlier in the same task, or Mem0 retrieval accuracy drops on long workflows.

2. Tool Routing Errors

The agent calls the wrong tool for the job — e.g., uses Brave Search when it should use Gmail, or invokes browser automation for something already in memory. Signals the tool list has grown too large for reliable LLM selection.

3. Cascading Task Failures

A single failing sub-task (e.g., browser automation timeout) causes the entire task to fail instead of just that branch. Signals the orchestration is too tightly coupled — sub-agents would isolate failures.

■ SOFT SIGNALS — Start Planning the Upgrade

4. Tool Count Crosses ~20

This is the practical ceiling where a single system prompt + tool list starts degrading LLM routing quality. Count your registered MCP tools. When you hit 20, start designing the sub-agent split — even if you haven't deployed it yet.

5. You Want Different Cost Tiers Per Task Type

Email classification should run on Haiku (\$0.0003/1K tokens). Complex research reasoning should run on Sonnet. If everything routes to Sonnet because orchestration is monolithic, you're overpaying 10-20x for simple tasks.

6. Response Latency Becomes Noticeable

A single agent loading the full memory context + all tool definitions + task history on every request gets slow. Sub-agents each carry only their relevant context — that's where you'll feel the performance difference.

7. You Want Different Personalities Per Domain

Email agent: concise and formal. Research agent: thorough and cited. Booking agent: confirmation-obsessed. A monolithic agent cannot hold multiple system prompt personalities simultaneously.

■ Metrics to Log From Day 1

Log these from the start — you need baseline data to know when thresholds are crossed.

Metric	Warning Threshold
Active tool count	> 20
Avg tokens per request	> 40K
Task success rate	< 90%
Wrong tool selections / day	> 3
Avg task completion time	> 30 seconds
Claude API cost / day	> \$3 (signals over-routing to Sonnet)

■ ■ What the Upgrade Looks Like When You Pull the Trigger

You won't rewrite anything. The current Agent Core becomes the Orchestrator Agent. You extract 4 sub-agents from it:

Sub-Agent	Tool Scope
Email Agent	Gmail tools only
Messaging Agent	WhatsApp / Telegram / SMS tools only
Research Agent	Brave / Firecrawl / Stagehand tools only
Booking Agent	Flight / restaurant / calendar tools only

Each sub-agent runs its own smaller system prompt and tool subset. The Orchestrator classifies the incoming task and hands off. This is a configuration change in OpenAI Agents SDK — not a structural rewrite.